

joker-lap Formula Reference

joker-lap — Stat Formula Quick Reference

Companion to CODEBASE.md Section 4/7. One row per formula instance found in the codebase — use this as the checklist when doing a formula-correction pass. Full code snippets and reasoning live in CODEBASE.md.

Batting stats

Stat	Location	Current formula	Correct MLB formula	Status
BA	All locations (<code>api/routers/mlb.py</code> : <code>batter_season_stats</code> , <code>lineup_stats</code> , <code>hitter_profile</code> , <code>lineup_batter_stats</code>)	<code>H / AB</code>	<code>H / AB</code>	✅ CORRECT
OBP	<code>batter_season_stats</code> (:2781-2783), <code>lineup_stats</code> (:2850-2852), <code>batter-vs-hand</code> <code>helper</code> (:2214-2215)	<code>(H + HBP) / (AB + HBP + SF)</code>	<code>(H + BB + HBP) / (AB + BB + HBP + SF)</code>	❌ WRONG — BB omitted entirely
OBP	<code>hitter_profile</code> (:2963-2967)	<code>(H + BB + HBP) / (AB + BB + HBP + SF)</code>	same	✅ CORRECT — use this as the template everywhere else
OBP	<code>hitter/{id}/juiced-splits</code> (:4352-4354, :4371-4377)	event-list AVG; denominator doesn't exclude sac bunts	same, but exclude SH from denominator	❌ WRONG (minor)
OBP	<code>game/{game_pk}/batter-splits/{id}</code> → <code>game_state_split</code> (:5141-5142)	event list omits Hit By Pitch	include HBP	❌ WRONG — inconsistent with sibling <code>ops_vs_hand</code> subquery (:5011-5019) in the same response
OBP	<code>lineup_batter_stats</code> (:5193-5195)	<code>(H + BB) / (AB + BB)</code> — no COALESCE on walks	<code>(H + BB + HBP) / (AB + BB + HBP + SF)</code>	❌ WRONG — HBP/SF omitted, NULL-unsafe
SLG	<code>batter_season_stats</code> , <code>lineup_stats</code> , <code>hitter_profile</code> , <code>lineup_batter_stats</code>	<code>TB / AB</code>	<code>TB / AB</code>	✅ CORRECT
SLG	<code>hitter/{id}/juiced-splits</code> (:4348-4351) + 4 more subqueries in <code>game/{game_pk}/batter-splits/{id}</code>	AVG(TB-per-event) over all qualifying rows incl. BB/HBP/SF/SH	<code>TB / AB</code> only	❌ WRONG — denominator too large, dilutes SLG
OPS	Everywhere	<code>obp + slg</code> (from same query's own OBP/SLG)	<code>OBP + SLG</code>	✅ mechanically correct, ❌ inherits any upstream OBP/SLG bug
PA	<code>hitter_profile</code> (:2951-2952)	<code>AB + BB + HBP + SF + SH</code>	<code>AB + BB + HBP + SF + SH</code>	✅ CORRECT
PA	Everywhere else (:3212, 5023, 5075, 5111, 5138)	<code>COUNT(*)</code> of <code>at_bats</code> rows excluding baserunning-only events	same as above	⚠️ reasonable proxy, not independently verified
K% (batter)	<code>batter_season_stats</code> (:2791), <code>lineup_stats</code> (:2860)	<code>K / AB</code>	<code>K / PA</code>	❌ WRONG — the classic K/AB-instead-of-K/PA error
K% (batter)	<code>hitter_profile</code> (:2975-2980)	<code>K / (AB+BB+HBP+SF+SH)</code>	<code>K / PA</code>	✅ CORRECT
K% (batter)	<code>game/{game_pk}/batter-splits/{id}</code> event-based (:5073-5074, 5146-5147)	<code>K / (qualifying at_bats rows)</code>	<code>K / PA</code>	✅ reasonable PA-proxy
BB% (batter)	(not found anywhere as an actual stat)	—	<code>BB / PA</code>	❑ MISSING FEATURE — only <code>proj_bb_pct</code> (projected) exists

Stat	Location	Current formula	Correct MLB formula	Status
OBP (simulation)	<code>simulate.py::aggregate_outcomes</code>	$(H+BB) / n_{sims}$	N/A — proprietary sim output, not a real-data stat	documented limitation: simulator doesn't model HBP as a distinct outcome
OBP (splits library)	<code>build_splits.py::split_stats()</code> (:75-97)	$(H+BB) / PA$ — no HBP/SF	$(H+BB+HBP) / (AB+BB+HBP+SF)$	WRONG (but note: this file isn't wired into the scheduler at all — see CODEBASE.md §3)
OBP (context splits)	<code>build_context_splits.py</code> (obp_against)	$(H+BB) / PA$ — no HBP/SF	same fix	WRONG
OBP (Juiced+ v1)	<code>build_goose.py::compute_juiced_scores</code>	hits / PA — BB and HBP omitted from numerator	$(H+BB+HBP) / PA$	WRONG — depresses Juiced+ v1 for high-walk batters; Juiced+2 (<code>build_goose2.py</code>) does not have this bug

Pitching stats

Stat	Location	Current formula	Correct MLB formula	Status
ERA	<code>lineup_pitcher_stats</code> (:5230-5232)	$ER \times 9 / \text{SUM}(\text{innings_pitched})$	$(ER / IP) \times 9$	WRONG IN PRACTICE — arithmetic is right, but <code>innings_pitched</code> input is corrupted (see IP conversion b)
ERA	Pitcher Profile / season-comparison / everywhere else	Pass-through from <code>mlb.fangraphs_pitching.era</code>	—	CORRECT (Fangraphs' own)
WHIP	Not computed locally anywhere	Pass-through from <code>mlb.fangraphs_pitching.whip</code>	$(BB+H) / IP$	CORRECT today, landr ever locally computed (would t bug)
K/9	Not found anywhere	—	$(K/IP) \times 9$	MISSING FEATURE
BB/9	Not found anywhere	—	$(BB/IP) \times 9$	MISSING FEATURE
K% (pitcher)	<code>build_daily_projections.py::get_pitcher_season_k_pct</code> (:185-199)	$K / (\text{qualifying at_bats rows})$	K / BF	reasonable BF-proxy (proje input only, not a displayed “act stat”)
IP conversion	<code>transform.py:349</code>	<code>float("6.1")</code> → 6.1 (literal decimal)	"6.1" → 6.333 (⅓), "6.2" → 6.667 (⅔)	WRONG — highest-impact the audit. Propagates incorrect <code>lineup_pitcher_stats</code> <code>mlb.daily_actuals.act</code> (<code>build_actuals.py:39</code>)
IP conversion	<code>build_daily_projections.py::get_pitcher_pa_per_inn</code> (:215-216)	$\text{FLOOR}(ip) + (ip - \text{FLOOR}(ip)) \times 10/3$	same	CORRECT — the one place done right
IP conversion (display)	<code>props_live</code> endpoint (:5692)	raw float cast of live-API IP string	N/A, display-only	minor — not aggregated, ju confusing display

Model metrics (proprietary — documented, not “graded” against an external formula)

Metric	Defined in	What it means	Notable finding
<code>proj_k_pct</code>	<code>build_daily_projections.py</code> (sim → lineup-rescale → season-blend, :372, 444-447, 452-459)	Per-PA (batters-faced) K rate, as a percent (24.5 = 24.5%)	Confirmed per-PA by its own downstream multiplication against <code>pa_per_inn</code>
<code>proj_ks_6inn</code>	<code>build_daily_projections.py:461</code>	$(\text{proj_k_pct}/100) \times \text{proj_ip} \times \text{pa_per_inn} \times \text{lineup_k}$	Name is misleading — <code>proj_ip</code> floats with the pitcher's real projected workload, not hardcoded to 6
<code>proj_ks_locked</code>	<code>build_daily_projections.py:544-546</code> , <code>upsert COALESCE</code> (:505-526)	Frozen snapshot set once a real posted lineup is	<code>build_actuals.py:48</code> falls back to live

Metric	Defined in	What it means	Notable finding
		used; permanent for the day thereafter	<code>proj_ks_6inn</code> if nothing was ever locked
K_PCT_CALIBRATION	<code>build_daily_projections.py:40</code> (fallback <code>0.830</code>) + <code>mlb.projection_calibration</code> (DB-driven)	Flat multiplier correcting the simulator's hot K% bias	Nightly-updated by <code>build_calibration.py</code> 's EWMA-style nudge (<code>LEARNING_RATE=0.30</code>)
<code>lineup_k_factor</code>	<code>build_daily_projections.py::get_lineup_k_factor</code> (:311-343)	Dampened (50%) adjustment for how K-prone the opposing lineup is vs. league (<code>0.224</code>)	✗ Its own SQL uses K/AB , while sibling functions in the same file use K/PA/BF — an internal inconsistency, not an external-formula bug
pHR+ “hard-hit” gate	<code>build_pk_phr.py:96</code>	Should gate on exit velocity	✗ WRONG — uses <code>start_speed >= 92</code> (pitch velocity), not <code>launch_speed >= 95</code> (exit velocity, used correctly everywhere else in the model family)
Team defense SLG allowed	<code>build_team_defense.py:70</code>	Should be <code>TB/AB</code>	✗ WRONG — denominator is PA (includes BB/HBP), not AB

Fix priority (see CODEBASE.md §7 for full detail on each)

1. 🚫 **IP notation bug** (`transform.py:349`) — corrupts ERA + `daily_actuals.actual_ip` + model-accuracy grading
2. 🚫 **OBP missing BB** (3 endpoints) — visible on Hitter Profile & lineup cards
3. 🚫 **SLG wrong denominator** (5 locations) — visible on Hitter Profile Juiced+2 splits & Game Companion
4. 🚫 **Batter K% = K/AB** (2 endpoints) — the textbook “common error”
5. 🟡 **pHR+ hard-hit gate uses pitch velocity** — corrupts a model blended into Goose+3
6. 🟡 OBP missing HBP (`lineup_batter_stats`)
7. 🟡 OBP missing HBP, inconsistent within one endpoint (`game_state_split`)
8. 🟡 Juiced+ v1 OBP omits BB/HBP (`build_goose.py`)
9. 🟡 `lineup_k_factor` K/AB vs. sibling K/PA inconsistency
10. 🟡 Team defense SLG-allowed divides by PA not AB
11. 🟡 OBP denominator includes sac bunts (`juiced-splits`)
12. 🟡 NULL-unsafe walks in `lineup_batter_stats` OBP
13. 🟢 `bapv_config.py` orphaned/unenforced
14. 🟢 `build_count_mix.py` not scheduled
15. 🟢 Missing BB%, K/9, BB/9